The empirical recurrence for OEIS A316806: recovering a conjecture that is not written down

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Abstract

OEIS A316806 records "Empirical recurrence of order 59", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 138$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A316806 is "Number of nX6 0..1 arrays with every element unequal to 0, 1, 2, 3, 4, 5 or 6 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$32, 2048, 113606, 6256258, 347046960, 19261290694, 1068884675914, 59316329979368, \dots$$

The entry states:

> Empirical recurrence of order 59 (see link above)

As of the "Last modified" line on the live entry (Jul 14 2018 14:59:05, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 138$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 138.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 276$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{59} c_i\, a(n-i),$$

c_{1}	54	c_{16}	−270993218675	c_{31}	−298360842697713	c_{46}	79337839312692
c_{2}	53	c_{17}	263684118878	c_{32}	−80351806902590	c_{47}	1535341687164
c_{3}	1700	c_{18}	−185817413420	c_{33}	−439672000435597	c_{48}	−52305100912624
c_{4}	544	c_{19}	−2598592097224	c_{34}	89707570372405	c_{49}	−3025148522320
c_{5}	−154733	c_{20}	7679580658180	c_{35}	795969436902831	c_{50}	5014639913088
c_{6}	−51449	c_{21}	1113240585822	c_{36}	40746507615022	c_{51}	9451770270912
c_{7}	−993511	c_{22}	−10257355732470	c_{37}	−74897242664050	c_{52}	5131486955648
c_{8}	4223723	c_{23}	38944791797825	c_{38}	−12159331770041	c_{53}	−3848169210368
c_{9}	83858362	c_{24}	−60719166283466	c_{39}	−541944767784342	c_{54}	−1912550585344
c_{10}	−280660965	c_{25}	−115812837144571	c_{40}	64242665789923	c_{55}	299116937216
c_{11}	−333665673	c_{26}	140959187632633	c_{41}	298604320174352	c_{56}	242684149760
c_{12}	2561179419	c_{27}	12994719021024	c_{42}	−136159535802902	c_{57}	127642402816
c_{13}	−7986871985	c_{28}	10014396319093	c_{43}	56249341684670	c_{58}	−5397020672
c_{14}	22106943997	c_{29}	340020442959978	c_{44}	25637740349170	c_{59}	−19084083200
c_{15}	52727543231	c_{30}	−122241753033475	c_{45}	−67752110677924		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59+4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.